

William Valentine

Stamford, CT | valentwa@rose-hulman.edu | (203) 391-8920

Education: **Bachelor of Science, Computer Science** **May 2027**

Rose-Hulman Institute of Technology, Terre Haute, IN

Minors: Cognitive Science, Mathematics, Imaging

Relevant courses: Human-Robot Interaction, Cognitive Psychology, Computer Vision, Deep Learning, Machine Learning, Linear Algebra, Artificial Intelligence, Deep Reinforcement Learning, Probability with Applications in Statistics

Skills: Programming Languages: JavaScript, Python, Java, C, Assembly, Scheme
Systems: Windows, Macintosh, Linux

Research: **Indiana University, Bloomington, IN** **Summer 2025 - Present**
Research Intern

- Worked alongside Dr. David Crandall and Dr. Selma Šabanović to create novel social robots (QTrobot, Baxter) for interacting with older adults
- Analyzed large-scale user studies with cross-cultural groups of older adults

University of Nevada, Reno, Reno, NV **Summer 2024 - Winter 24**
REU Site: Collaborative Human-Robot Interaction for Robots in the Field

- Worked alongside Dr. David Feil-Seifer and Dr. Emily Hand to create one of the first systems for the detection of human comfort and discomfort
- Used deep-learning powered pose estimation, valence and arousal, audio transcription, and textual sentiment analysis to model human behavior

Rose-Hulman Institute of Technology, Terre Haute, IN

Behavior Analysis for Adaptive Robotic Behavior **Winter 25 - Present**

- Worked alongside Dr. Michael Wollowski to explore how real-time video-based behavior analysis could be used to create robots with adaptive behavior

Resolution of command ambiguity using LLMs for robots **Fall 2024 - Fall 25**

- Worked alongside Dr. Michael Wollowski to introduce a system and benchmark for resolving confusing parts of commands given to robots using images

LiDAR Point Cloud Alignment Using Log-Gabor Filters **Fall 23 - Summer 24**

- Worked alongside Dr. Lixing Song to address alignment issues caused by learning based point cloud alignment methods for self-driving vehicles
- Successfully created an algorithm for aligning point clouds in complex, uncertain environments

Vanderbilt University, Nashville, Tennessee **Summer 2025**
VORTEX (Vanderbilt Online Research Training and Exploration) Member

- Invited to join the first cohort of a summer program focusing on research ethics and research writing

Publications Under Review or Revision: (conferences obscured for blind review):

Valentine, W., Kamino, W., Hsu, L., Sabanovic, S., Crandall, D., & Khoo, W. (2026). *Novel approach to in-the-wild audio-visual datasets for emotion recognition, Computer Vision Conference.*

Valentine, W., Wang, Y., Kamino, W., Sato, H., Swaminathan, M., Noguchi, A., Ramirez, F., Hirata, Y., Oh, J., Nagata, S., Tsui, K., Hsu, L., Sabanovic, S., Crandall, D., & Khoo, W. (2026). *Large-scale cross-cultural user study of social robots interacting with older adults, Robotics Journal.*

Publications:

Valentine, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). *BehaviorKit: A Multi-modal Real-time Behavior Analysis Library for Robots, HRI 2026 (Short paper, Acceptance Rate: 38%)*

Valentine, W., Wollowski, M. (2025). *Assessment of the capabilities of multimodal large language models in locating and resolving ambiguities during human-robot teaming. AHFE 2025.*

Valentine, W., Webb, M., Collum, C., Feil-Seifer, D., & Hand, E. (2024). *HCC: An explainable framework for classifying discomfort from video. ISVC 2024 (Oral: 45.1%).*

Valentine, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). *From Long-Term HRI to Dementia Care: Scoping Personalization Approaches, HRI 2026 Workshop: LEAP-HRI*

Song, L., **Valentine, W.**, Yang, Q., Wang, H., Fang, H., & Liu, Y. (2024). *BB-Align: A lightweight pose recovery framework for vehicle-to-vehicle cooperative perception. ICDCS 2024 (21.9%).*

Connor B., **Valentine, W.**, Hsu, L., Kamino, W., Šabanović, S., Crandall, D., & Khoo, W. (2026). *Ghost in the Machine: Exploring Designed Fear in HRI, HRI 2026 LBR*

Khoo, W., Hsu, L., **Valentine, W.**, Connor, B., Nil, T., Khoo, A., Kamino, W., Šabanović, S., Crandall, D. (2026), *Meet IRIS: An Interactive Robotic Experience for Exploring Life Purpose, HRI 2026 Demo*

Experience: **Grader and TA, CSSE Department, RHIT** **Spring 2024 - Present**

- Graded for Intro to Software Development, Web Development, Programming Language Concepts, Data Structures and Algorithms, and Mechatronics
- Head TA for Programming Language Concepts: Winter 2024, 2025
- Created and designed an automatic grading system utilizing Python

Service: HRI Main Track 2026 [3 reviews]
HRI Short Papers Track 2026 [1 review]
HRI Late-Breaking Report Track 2026 [3 reviews]
CHI Late-Breaking Work Track 2026 [3 reviews]
AAAI: AI Magazine [2 reviews]
AAAI: AIES SP 2025 [2 reviews]
ICDCS Student Volunteer

Honors and Roles: **Rose-Hulman Institute of Technology, Terre Haute, IN**
Rose Research Fellow, Nominated for CSSE TA of the year
Rose Research Fellow Mentor

Goldwater Scholar Nominee 2025-2026
CSSE Student Advisory Board Member
Academic Computing Implementation Committee (Student Rep)

HRI2026: Special Recognition for Outstanding Reviews

Grants: **National Science Foundation, Alexandria, Virginia**

- Conference Travel Award \$500

Summer 2024

Rose-Hulman Institute of Technology, Terre Haute,

- Rose Research Fellows \$500, \$1000
- IN IP/ROP 2024 \$500
- CSSE Departmental \$2000, \$1000

Fall 2024, Spring 2025

Spring 2024

Spring 2024, 2025